DTC P0101:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC P0101:00	MAF sensor circuit range/performance problem
DETECTION CONDITION	• The difference between the intake air amount measured by the MAF sensor and the estimated intake air amount estimated by the MAP sensor is outside of the specified value. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition in two consecutive drive cycles or in one drive cycle while the DTC for the same malfunction has been stored in the PCM. • PENDING CODE is available if the PCM detects the above malfunction condition during first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• MAP sensor/IAT sensor No.2 connector or terminals malfunction • MAF sensor/IAT sensor No.1 connector or terminals malfunction • MAP sensor/IAT sensor No.2 loose • MAF sensor/IAT sensor No.1 loose • PCM connector or terminals malfunction • MAP sensor malfunction • MAF sensor malfunction • Air leakage from intake-air system • Purge solenoid valve malfunction • PCV valve malfunction • A non-genuine air cleaner or air cleaner cover installed • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**VERIFY CURRENT INPUT SIGNAL STATUS IS CONCERN INTERMITTENT OR CONSTANT** • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Start the engine. • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) — ECT — TP_REL — RPM • Warm up the engine until the ECT PID is **above 70 °C {158 °F}**. • Perform the following: 1. Depress the accelerator pedal to increase the engine speed to **approx. 5,000 rpm**. 2. Release the accelerator pedal to decrease the engine speed to idle speed. 3. Repeat Step 1 and Step 2 operations above **3 times** in succession. • Perform the Pending Trouble Code Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the PENDING CODE for this DTC present?	Yes	Go to the next step.
		No	Intermittent concern exists. • Perform the “INTERMITTENT CONCERN TROUBLESHOOTING” procedure. (See INTERMITTENT CONCERN TROUBLESHOOTING [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4	**INSPECT MAP SENSOR/IAT SENSOR NO.2 CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the MAP sensor/IAT sensor No.2 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
5	**INSPECT MAF SENSOR/IAT SENSOR NO.1 CONNECTOR CONDITION** • Disconnect the MAF sensor/IAT sensor No.1 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
6	**INSPECT INSTALLATION OF MAP SENSOR/IAT SENSOR NO.2** • Inspect installation of MAP sensor/IAT sensor No. 2. • Is the MAP sensor/IAT sensor No.2 installed securely?	Yes	Go to the next step.
		No	Retighten the MAP sensor/IAT sensor No.2, then go to Step 14. (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.2 REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
7	**INSPECT INSTALLATION OF MAF SENSOR/IAT SENSOR NO.1** • Inspect installation of MAF sensor/IAT sensor No. 1. • Is the MAF sensor/IAT sensor No.1 installed securely?	Yes	Go to the next step.
		No	Retighten the MAF sensor/IAT sensor No.1, then go to Step 14. (See MASS AIR FLOW (MAF) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
8	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**INSPECT MAP SENSOR** • Reconnect all disconnected connectors. • Inspect the MAP sensor. (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the MAP sensor/IAT sensor No.2, then go to Step 14. (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.2 REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
10	**INSPECT MAF SENSOR** • Inspect the MAF sensor. (See MASS AIR FLOW (MAF) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the MAF sensor/IAT sensor No.1, then go to Step 14. (See MASS AIR FLOW (MAF) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
11	**INSPECT INTAKE-AIR SYSTEM FOR AIR LEAKAGE** • Inspect for leakage in intake-air system. • Is there any leakage?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 14.
		No	Go to the next step.
12	**INSPECT PURGE SOLENOID VALVE** • Inspect the purge solenoid valve. (See PURGE SOLENOID VALVE INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the purge solenoid valve, then go to Step 14. (See PURGE SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
13	**INSPECT PCV VALVE OPERATION** • Inspect the PCV valve operation. (See POSITIVE CRANKCASE VENTILATION (PCV) VALVE INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the PCV valve, then go to the next step. (See POSITIVE CRANKCASE VENTILATION (PCV) VALVE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Replace with a genuine air cleaner or air cleaner cover if a non-genuine part is installed. Go to the next step.
14	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Start the engine. • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) — ECT — TP_REL — RPM • Warm up the engine until the ECT PID is **above 70 °C {158 °F}**. • Perform the following: 1. Depress the accelerator pedal to increase the engine speed to **approx. 5,000 rpm**. 2. Release the accelerator pedal to decrease the engine speed to idle speed. 3. Repeat Step 1 and Step 2 operations above **3 times** in succession. • Perform the Pending Trouble Code Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the PENDING CODE for this DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
15	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.